motion of an electron. The best we can do is to try to understand electrons and their motion as mathematical abstractions.

"So what?" says the skeptic. "Classical mechanics is filled to the brim with mathematical abstractions—point masses, rigid bodies, inertial reference frames, positions, momenta, fields, waves—the list goes on and on. There's nothing new about mathematical abstractions." This is actually a fair point, and indeed the classical and quantum worlds have some important things in common. Quantum mechanics, however, is different in two ways:

1. *Different Abstractions*. Quantum abstractions are fundamentally different from classical ones. For example, we'll see that the idea of a state in quantum mechanics is conceptually very different from its classical counterpart. States are represented by different mathematical objects and have a different logical structure.

2. *States and Measurements*. In the classical world, the relationship between the state of a system and the result of a measurement on that system is very straightforward. In fact, it's trivial. The labels that describe a state (the position and momentum of a particle, for example) are the *same* labels that characterize measurements of that state. To put it another way, one can perform an experiment to determine the state of a system. In the quantum world, this is not true. States and measurements are two different things, and the relationship between them is subtle and nonintuitive.

These ideas are crucial, and we'll come back to them again and again.

1.2 SPINS AND QUBITS

The concept of spin is derived from particle physics. Particles have properties in addition to their location in space. For example, they may or may not have electric charge, or mass. An electron is not the same as a quark or a neutrino. But even a specific type of particle, such as an electron,